Operating Systems & DBMS Technical Interview Capsule

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1. OPERATING SYSTEMS CORE INTERVIEW QUESTIONS

• **Process vs Thread:** A process has its own private virtual memory space (code, data, heap). Threads share the memory space of their parent process but maintain their own program counter, registers, and call stack.
• **Deadlock 4 Coffman Conditions:** 1. Mutual Exclusion; 2. Hold and Wait; 3. No Preemption; 4. Circular Wait. Breaking any single condition prevents deadlock. Banker's algorithm checks safe states.
• **Virtual Memory & Thrashing:** When high page fault frequency causes CPU to spend more time swapping pages between disk and RAM than executing instructions. Solved by increasing RAM or reducing degree of multiprogramming.

2. DATABASE MANAGEMENT SYSTEMS (DBMS)

• **ACID Properties:** Atomicity (all or nothing, WAL undo logs), Consistency (schema integrity rules), Isolation (locking / MVCC), Durability (WAL redo logs persisted to disk).
• **B+ Tree Indexing:** All records stored in leaf nodes linked sequentially for range queries. Internal nodes only store keys for navigation. Balanced tree with guaranteed O(log N) lookup, insertion, and deletion.
• **SQL Normalization (1NF to BCNF):** 1NF: Atomic values. 2NF: 1NF + no partial dependency on composite key. 3NF: 2NF + no transitive dependency (non-prime -> non-prime). BCNF: For every functional dependency X -> Y, X must be a super key.